

Nathan Genetzky's Resume

Software and Hardware

Nathan Genetzky

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Career Profile

- Embedded Linux engineer with ~9 years taking connected products from board bring-up through secure, field-updatable deployment.
- Yocto distributions, BSP and kernel work, CI/CD, and hardware-in-the-loop test.
- Product cybersecurity (EN 18031 / EU RED, secure boot, SBOM) and OTA update systems.
- Electrical Engineering graduate with honors distinction.

Contact

- nathan@genetzky.us
- academic.genetzky.us
- linkedin.com/in/genetzky
- github.com/NGenetzky
- twitter.com/ngenetzky
- keybase.io/ngenetzky

Experiences

Senior Software Engineer

PTx, AGCO's Precision Agriculture Technology business (04.2024 - PRESENT)

- Develop embedded Linux platforms for agricultural equipment electronics; working knowledge of CAN bus and protocol.
- Applied EN 18031 (EU RED cybersecurity) to a connected product: asset identification, decision-tree assessment, and evidence for third-party certification; tracking CRA readiness.
- Implemented secure boot and signed firmware images; performed threat modeling and key management backed by a hardware security element.
- Generate and maintain SBOMs for a Yocto-based product.
- Implemented RAUC OTA updates across multiple A/B and asymmetric partition layouts: U-Boot integration, secure boot fallback handling, recovery from failed kernel, rootfs, and application updates, and the tests to prove it.
- Build and deliver containerized applications that run on embedded targets.
- Integrated AI-assisted development into daily embedded workflow for code generation, review, test authoring, and documentation, with defined boundaries on where generated code is not trusted without verification.

Senior Software Engineer

ESW Team at Appareo Systems (acquired by AGCO, 01.2022) (12.2021 - 04.2024)

- Performed board bring-up for custom PCB with heterogeneous SOC (IMX8DXL), debugging with JTAG/SWD, GDB, logic analyzer, and oscilloscope.
- Integrated drivers for cellular modems, WiFi, and bluetooth.
- Setup dockerized Jenkins with manual and GitOps centric builds, using Gitflow with automated build and deployment of releases.
- Developed and utilized HIL testing with Labgrid on RPI4.
- Generated SDKs and docker images for quick cross-compilation of C++ Applications.

Software Engineer 1

System Team at Vaddio (05.2017 - 12.2021)

- Working with FPGA and python developers to create an integrated embedded system that routes audio and video signals in professionally built AV systems.
- I help to build a custom Linux distribution from the 'up' with the Yocto Project.

- Our team uses many other tools during development such as Jenkins, Jira, Git, Bitbake, and Puppet.

Software Engineer

Dojo Five (11.2018 - 12.2021)

- Develop embedded software for Particle, Nordic, and STM32 microcontrollers
- Utilized custom hardware with the Particle platform for a custom door access solution
- Developed with NRF52 SDK to create devices in a complex robotic system with BLE services

Embedded Linux and OTA Experiments

Personal projects, self-directed (ONGOING)

- Compared Mender and OSTree against RAUC to understand A/B, atomic, and rollback-safe update strategies, and authored a custom update agent to work through the trade-offs first hand.
- Managed device fleets with Balena: provisioning, remote access, and delivery of containerized applications to embedded targets.

Projects

Hardware Integrated Prototyping Environment at SDSU Engineering Expo

- Nathan Genetzky, Jordan Ulmer, Tanner Johnson (SDSU, 2017)
- <http://academic.genetzky.us/project/sdsu/2017-hipe/>

Design and Verification of a SPI to JTAG Interface Adapter

- Nathan Genetzky, Jordan Ulmer (SDSU, 2017)
- <http://academic.genetzky.us/publication/sdsu/2017-05-05-ee492-spi2jtag/>

Interactive User Interface with PIC18 Microcontroller

- Nathan Genetzky, Drake Jenö (SDSU, 2017)
- <http://academic.genetzky.us/publication/sdsu/2015-05-01-ee347-interactive-ui-with-pic-microcontroller/>

Particle Projects

- Firmware for micro controllers sold by particle.io for use with wifi or cellular cloud devices.
- <http://academic.genetzky.us/project/particle-projects/>

Skills & Proficiency

- Yocto (Build System, Embedded Distro, BSP Support) (90%)
- Developer Tools (Docker, Jenkins, Git, GDB, JTAG/SWD) (85%)
- Terminal Workflow (Vim, GNU tools, shell scripting) (85%)
- Linux Kernel (Drivers, Config, BSP) (55%)
- Python Middleware (DBus, REST, GObject, threading) (35%)
- FPGA Development (SW Interface, Reusable Verilog, Automated Builds) (25%)

Education

BS Electrical Engineering South Dakota State University (2012 - 2017)

Minor Software Engineering South Dakota State University (2012 - 2017)

Lauguages

- Bash (~10 years)
- Python (~8 years)
- C++ (~9 years)
- C (~7 years)
- Dockerfile (~5 years)
- Java for Android (~1 year)

Interests

- hobby-electronics, home automation, embedded-linux
- ice and rock climbing
- snowboarding and skiing
- gardening, cooking
- jetskiing, boating, water skiing
- camping, kayaking, volleyball